SHIPMENT AND BOOKING ANALYTICS DASHBOARD

Using Microsoft Power BI

Project Report

ABSTRACT

This project presents the design and development of an interactive analytics dashboard using Microsoft Power BI to analyze shipment operations and customer booking information. The work follows a structured workflow beginning with data collection and preparation, followed by data modeling, and concluding with the creation of interactive visualizations.

The dashboard integrates two primary datasets — Booking Data and Shipment Data — to provide a unified view of logistics performance and customer behavior. Visualizations include geographic maps, bar charts, gauge indicators, donut charts, and KPI cards, enabling stakeholders to explore shipment distribution, shipping costs, customer satisfaction, and product category demand.

The results demonstrate how business intelligence tools can transform raw operational data into meaningful insights that support decision-making and improve logistics efficiency.

TABLE OF CONTENTS

Abstract i

CHAPTER 1 .....
Introduction 1

CHAPTER 2 .....
Tools and Technologies Used 3

CHAPTER 3 .....
Dataset Description and Data Preparation 5

CHAPTER 4 .....
Dashboard Visualizations 8

CHAPTER 5 .....
Results, Observations, and Future Scope 11

Conclusion 13

CHAPTER 1
INTRODUCTION

Business Intelligence (BI) tools play a vital role in helping organizations convert raw data into meaningful insights that support informed decision-making. In today's data-driven environment, logistics and e-commerce companies generate large volumes of operational data related to shipments, customer bookings, and service usage. Effectively analyzing this data is essential for improving operational efficiency and understanding customer behavior.

In this project, Microsoft Power BI was used to analyze shipment operations and customer booking information. Power BI is a widely adopted BI platform that enables users to connect to multiple data sources, transform and model data, and build interactive visualizations through an intuitive interface.

The primary objective of this project is to develop a clear and interactive dashboard that provides insights into shipment performance, customer preferences, and service trends. The dashboard integrates two datasets — Booking Data and Shipment Data — and presents the findings through a variety of visualizations including maps, bar charts, gauges, donut charts, and KPI indicators.

1.1 Motivation

Manual analysis of large shipment and booking datasets is time-consuming and prone to errors. Business intelligence dashboards address this challenge by automating data aggregation and presenting information in a visual format that allows users to identify trends, compare metrics, and make data-driven decisions quickly. This project was motivated by the need to demonstrate how Power BI can be used to build such a solution for a logistics and customer booking use case.

1.2 Project Objectives

The following objectives were defined for this project:

- Analyze shipment distribution across different countries and states using geographic visualization.
- Evaluate shipping cost trends based on shipment modes such as First Class, Second Class, and Standard Class.

- Examine customer booking behavior and service preferences.
- Study customer review ratings and feedback to assess satisfaction levels.
- Visualize product category demand and corresponding shipment quantities.
- Develop an interactive Power BI dashboard for self-service data analysis.

1.3 Scope of the Project

The scope of this project is limited to data visualization and analytical modeling using Power BI. The project does not involve real-time data integration or deployment on a production server. All analysis is performed on static datasets that were cleaned and structured prior to import into Power BI.

CHAPTER 2
TOOLS AND TECHNOLOGIES USED

This chapter describes the tools and technologies used at each stage of the project. The development of the analytics dashboard was carried out using Microsoft Power BI as the primary platform, supported by its built-in data transformation and modeling features.

2.1 Tools Used

The following tools were used during the implementation of the project:

- – Microsoft Power BI Desktop
  Used for data visualization, dashboard design, and interactive report creation.
- – Power Query Editor
  Used for data ingestion, cleaning, and transformation prior to visualization.
- – DAX (Data Analysis Expressions)
  Used for writing calculated measures and custom metrics within the data model.
- – Data Modeling (Power BI)
  Used to define relationships between datasets for integrated cross-dataset analysis.

2.5 Workflow Summary

The complete workflow of the project can be summarized as follows:

1. Import of raw datasets into Power BI Desktop
2. **Power Query Editor**Data cleaning and transformation using
3. Data modeling to establish relationships between tables
4. **DAX**Creation of calculated measures using
5. Design of interactive visualizations and dashboard layout

Each step was carried out sequentially to ensure data quality and consistency throughout the development process.

CHAPTER 3
DATASET DESCRIPTION AND DATA PREPARATION

This chapter describes the two datasets used in this project and the data preparation steps applied before visualization. The datasets were imported into Power BI and linked through a relational data model.

3.1 Overview of Datasets

Two datasets were used in this project:

- Booking Data – Contains customer booking details, service fees, review ratings, and cancellation policies.
- Shipment Data – Contains order and logistics information including product categories, shipping modes, costs, and sales values.

These datasets were integrated in Power BI to enable combined analysis of shipment activity and customer booking behavior.

3.2 Booking Data

The Booking Data dataset captures customer interaction and transactional details related to service bookings. It provides insight into customer preferences, satisfaction levels, and product popularity.

The dataset contains the following attributes:

- Instant Bookable – Indicates whether a booking was confirmed instantly or required manual approval.
- Cancellation Policy – The cancellation rule applied to the booking: Flexible, Moderate, or Strict.
- Service Fee – The fee charged to the customer for the booking service.
- Number of Reviews – Total count of customer reviews submitted.
- Review Rate Number – Numerical rating assigned by the customer.
- Rating Type – Categorical classification of the rating: Good, Excellent, or Bad.
- Popularity of Product – Indicator of product demand: In-Demand or Unpopular.

This dataset supports analysis of customer booking behavior, service fee trends, customer satisfaction levels, and product popularity.

3.3 Shipment Data

The Shipment Data dataset contains detailed order fulfillment and logistics information. It supports analysis of operational performance, cost management, and geographic distribution of shipments.

The dataset contains the following attributes:

- Category – Product category such as Furniture or Office Supplies.
- City, State, Country – Geographic location of the shipment destination.
- Customer Name – Name of the customer placing the order.
- Discount – Discount percentage applied to the order.
- Order Date – Date on which the order was placed.
- Order Priority – Priority level: Critical, High, Medium, or Low.
- Quantity – Number of units ordered.
- Sales – Total sales value for the order.
- Segment – Customer segment: Consumer, Corporate, or Home Office.
- Ship Date – Date on which the order was dispatched.
- Ship Mode – Shipment method: First Class, Second Class, or Standard Class.
- Shipping Cost – Cost incurred for shipping the order.
- Average Shipping Cost – Calculated DAX measure representing the average cost across shipments.
- Total Sales – Calculated DAX measure representing aggregated sales value.
- Year of Purchase – Derived field extracted from the Order Date for time-series analysis.

3.4 Data Preparation

Data preparation was performed using Power Query Editor in Power BI Desktop. The following transformation steps were applied to clean and structure the data for visualization:

- Removed redundant and unnecessary columns that were not relevant to the analysis.
- Renamed columns to improve readability and maintain consistent naming conventions.
- Converted data types to ensure correct aggregations and calculations.
- Extracted the year component from the Order Date field to support time-series analysis.
- Created calculated fields and derived attributes using DAX expressions.
- Validated and standardized categorical values to eliminate inconsistencies.

These steps ensured that the datasets were properly structured and ready for data modeling and visualization.

3.5 Data Modeling

The project uses a relational data model consisting of the two core tables: Booking Data and Shipment Data. Relationships were established between these tables in Power BI to enable integrated cross-dataset querying and filtering.

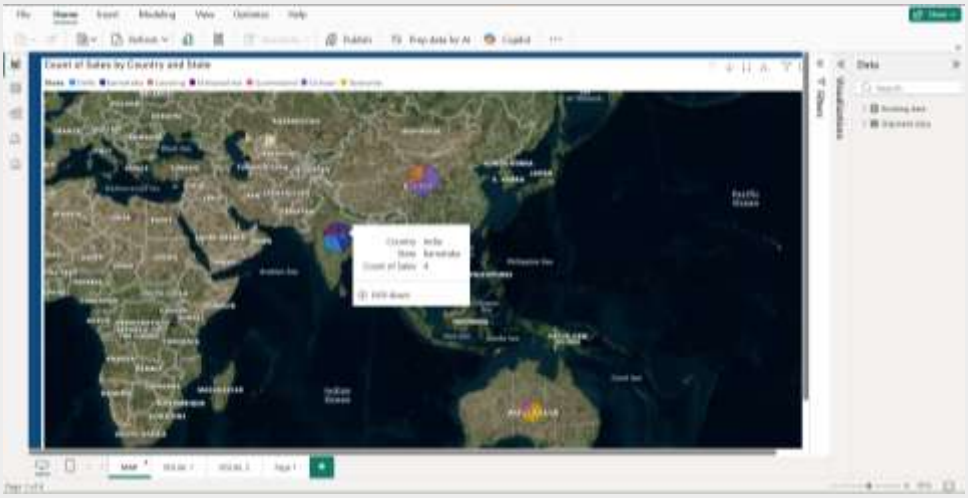
This model architecture allows Power BI to generate combined insights by linking customer feedback and booking metrics with shipment performance data, providing a complete picture of both operational and customer-facing dimensions.

CHAPTER 4
DASHBOARD VISUALIZATIONS

This chapter describes the visualizations developed in the Power BI dashboard. Each visualization was designed to surface specific insights from the integrated datasets. The dashboard allows users to interact with the data through filters, slicers, and drill-through options.

4.1 Geographic Shipment Analysis

A map visualization displays the distribution of sales across countries and states, helping identify regions with higher shipment activity.



4.2 Sales and Service Fee Analysis

A bar chart shows the number of sales and service fees per year, used to analyze yearly sales patterns and service charges.

4.3 Shipping Cost by Shipment Mode

A horizontal bar chart compares shipping costs across First Class, Second Class, and Standard Class shipment modes.

4.4 Customer Review KPI Card

A KPI card displays the average customer review rating to evaluate overall customer satisfaction.

4.5 Order Priority Indicator

A card visualization highlights the first order priority, helping understand the urgency level of shipments.

4.6 Total Quantity Shipped

A card visualization shows the total quantity of products shipped across all orders.

4.7 Customer Preference Gauge

A gauge visualization represents the number of customers according to their booking preference.

4.8 Instant Booking Review Flow

A flow diagram visualizes customer reviews for instantly booked orders compared to non-instant bookings.

4.9 Product Category Distribution

A donut chart shows the quantity shipped across product categories such as Furniture and Office Supplies.

CHAPTER 5
RESULTS, OBSERVATIONS, AND FUTURE SCOPE

This chapter presents the results obtained from the Power BI dashboard analysis and discusses key observations. It also outlines the limitations of the current implementation and potential areas for future development.

5.1 Results from Dashboard Analysis

The following results were observed from the interactive dashboard:

- The geographic map confirmed that shipment activity is concentrated in specific regions, indicating key markets for logistics focus.
- The bar chart showed that sales values varied across years, with certain periods showing higher transaction volumes.
- The shipping cost comparison revealed that Second Class shipment is the most frequently used mode, followed by Standard Class and First Class.
- The KPI card displayed an average customer review rating of approximately 3.5, indicating moderate satisfaction levels.
- The instant booking flow diagram showed that non-instant bookings are more common, suggesting limited adoption of the instant booking feature.
- The donut chart confirmed that Furniture and Office Supplies are the dominant product categories by shipment volume.

5.2 Observations

The following observations were made based on the analysis:

- The separation of booking and shipment data into distinct tables, connected through a relational model, enabled effective cross-dataset filtering and analysis.
- DAX-calculated measures such as Total Sales and Average Shipping Cost provided more meaningful metrics compared to raw data fields.
- The dashboard design demonstrated that a combination of different visualization types — maps, charts, gauges, and KPI cards — provides a more complete view of operations than any single chart type.
- An average review rating of 3.5 suggests that customer satisfaction is moderate and that there are areas for service improvement.
- The concentration of Second Class shipments suggests that customers prefer a balance between delivery speed and cost.

5.3 Limitations

The following limitations apply to the current implementation:

- The analysis is based on static datasets and does not support real-time data updates.
- The dashboard does not include predictive analytics or forecasting capabilities.
- No external data sources such as live APIs or databases were connected in this version.

5.4 Future Scope

The project can be extended in the following ways:

- Integration with live databases or APIs to enable real-time dashboard updates.
- Incorporation of predictive models to forecast demand, shipping costs, or customer churn.
- Addition of drill-through pages for detailed customer or product-level analysis.
- Deployment of the dashboard on the Power BI service for team-wide access and collaboration.
- Inclusion of additional datasets such as supplier data or customer demographics to enrich the analysis.

CONCLUSION

This project successfully demonstrates the use of Microsoft Power BI to analyze shipment and booking data and generate meaningful business insights. The dashboard integrates two datasets through a relational data model and presents the findings through a range of interactive visualizations.

The analysis confirmed that shipment activity is geographically concentrated, Second Class is the most used shipping mode, customer satisfaction is moderate, and Furniture and Office Supplies dominate shipment volumes. These insights can help organizations make informed decisions about logistics planning, cost management, and service improvement.

The project highlights the importance of business intelligence tools in transforming raw operational data into actionable insights. It also establishes a foundation for future work involving real-time data integration, predictive analytics, and broader deployment on collaborative BI platforms.

Submitted by

NAME: P Mahema Sai

STATUS: Student

COLLEGE: PES University , Bengaluru, Karnataka